MECHANICAL PROPERTIES OF SOLIDS

NCERT Class 11 Physics | JEE Main / NEET Formula & Method Sheet

Theme: Scientific Navy & Teal
Format: High-Density Revision

1. Stress, Strain & Hooke's Law

Fundamentals

LONGITUDINAL STRESS & STRAIN

MUST KNOW

$$\text{Tensile / Compressive Stress } (\sigma) = \frac{F}{A}, \quad \text{Longitudinal Strain } (\epsilon) = \frac{\Delta L}{L}$$

Symbols: F: Applied normal force (N), A: Cross-sectional area (m^2), L: Original length, ΔL: Elongation/compression.

SI Units: Stress: $\mathrm{N\,m^{-2}}$ or Pascal (Pa); Strain: Dimensionless [$M^0L^0T^0$].

Conditions: Force F must act perpendicular (normal) to the cross-sectional area A.

SHEARING & HYDRAULIC (VOLUME) STRAIN

MUST KNOW

$$\text{Shearing Strain } (\theta) = \frac{\Delta x}{L} = \tan\theta \approx \theta, \quad \text{Volume Strain} = \frac{\Delta V}{V}$$

Symbols: Δx: Lateral displacement of top face, L: Distance between opposite fixed faces, ΔV: Change in volume, V: Original volume.

Shear Stress: $\sigma_s = \frac{F_\parallel}{A}$, where $F_\parallel$ is tangential force applied parallel to the surface.

HOOKE'S LAW

MUST KNOW EXAM FAVOURITE

$$\text{Stress} \propto \text{Strain} \implies \text{Stress} = k \times \text{Strain} \implies k = \frac{\text{Stress}}{\text{Strain}}$$

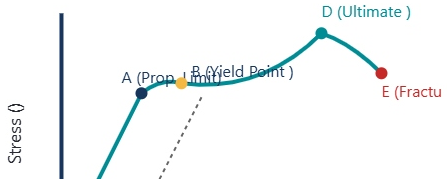
Modulus (k): Proportionality constant known as **Modulus of Elasticity**. Depends on material nature and temperature, independent of dimensions.

Validity: Strictly valid only within the proportional limit / small deformations ($\epsilon < 1\%$).

2. Stress-Strain Curve & Elasticity Analysis

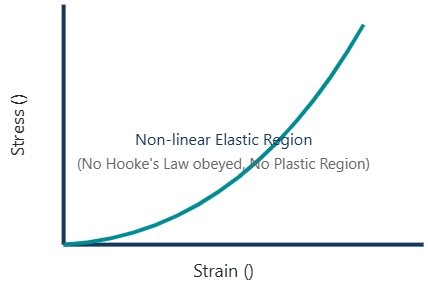
Graphical Analysis

A. Typical Metal Stress-Strain Curve





B. Elastomer Curve (Aorta Tissue)



⚠ EXAM CAUTION & INTERPRETATION OF STRESS-STRAIN GRAPHS:

1. **Ductile vs. Brittle Materials:** If fracture point E is far from ultimate strength D, material is **Ductile**. If D and E are close, it is **Brittle**.
2. **Young's Modulus from Slope:** $Y = \text{Slope of linear region } (OA) = \frac{\Delta\sigma}{\Delta\epsilon}$. Higher slope $\implies$ More rigid / elastic material (e.g., $Y_{\text{steel}} > Y_{\text{copper}}$).

3. Three Elastic Moduli (Young's, Shear, Bulk)

Core Formulations

1. YOUNG'S MODULUS (Y) — LENGTH CHANGE

MUST KNOW EXAM FAVOURITE

$$Y = \frac{\text{Tensile or Compressive Stress}}{\text{Longitudinal Strain}} = \frac{F/A}{\Delta L/L} = \frac{F \cdot L}{A \cdot \Delta L}$$

For Circular Wire: $A = \pi r^2 = \frac{\pi d^2}{4} \implies Y = \frac{4FL}{\pi d^2 \Delta L} \implies \Delta L = \frac{4FL}{\pi d^2 Y}$

Key Property: Defined **ONLY FOR SOLIDS**. Steel is more elastic than copper/rubber because it requires greater force for the same strain.

2. SHEAR MODULUS (G OR η) — SHAPE CHANGE

MUST KNOW

$$G = \frac{\text{Shearing Stress}}{\text{Shearing Strain}} = \frac{F_\parallel/A}{\Delta x/L} = \frac{F_\parallel/A}{\theta} = \frac{F_\parallel}{A \cdot \theta}$$

Empirical Relation: For most solids, Shear Modulus is approximately one-third of Young's Modulus: $G \approx \frac{Y}{3}$.

Applicability: Defined **ONLY FOR SOLIDS** (Fluids cannot sustain static shearing stress).

3. BULK MODULUS (B) & COMPRESSIBILITY (K) — VOLUME CHANGE

MUST KNOW EXAM FAVOURITE

$$B = -\frac{p}{\Delta V/V} \quad \text{Bulk modulus } B = -\frac{\Delta p}{\Delta V / V} = -V\frac{\Delta p}{\Delta V}, \quad \text{Compressibility } k = \frac{1}{B} \quad \text{Compressibility k} = \frac{1}{B} = -\frac{1}{V}\frac{\Delta V}{\Delta p}$$

Negative Sign: Indicates that increase in pressure ($p > 0$) causes decrease in volume ($\Delta V < 0$), keeping $B > 0$.

Fluid Depth Form: Pressure at depth h: $p = h\rho g \implies \frac{\Delta V}{V} = \frac{h\rho g}{B}$.

States Comparison: $B_{\text{solid}} > B_{\text{liquid}} > B_{\text{gas}}$. Gases are $\sim 10^6$ times more compressible than solids.

Modulus	Stress Type	Strain Type	Formula	Applicable State
Young's (Y)	Normal (F/A)	Longitudinal ($\Delta L/L$)	$Y = \frac{FL}{A\Delta L}$	Solids only
Shear (G)	Tangential ($F_{\parallel}/A$)	Shear angle ($\theta = \Delta x/L$)	$G = \frac{F_{\parallel}}{A\theta}$	Solids only
Bulk (B)	Hydraulic (p)	Volume ($\Delta V/V$)	$B = -\frac{pV}{\Delta V}$	Solids, Liquids, Gases

4. Poisson's Ratio & Elastic Potential Energy

Derived Properties

POISSON'S RATIO (ν OR σ)

MUST KNOW SPECIAL CASE

$$\nu = -\frac{\text{Lateral Strain}}{\text{Longitudinal Strain}} \quad \text{Poisson ratio } \nu = -\frac{\Delta d/d}{\Delta L/L} = -\frac{\Delta r/r}{\Delta L/L}$$

Unit & Dimension: Pure numerical ratio. Dimensionless $[M^0L^0T^0]$.

Theoretical Limit: $-1 \le \nu \le 0.5$. Practical value for engineering metals: $0.28 \le \nu \le 0.33$.

Derivation: Strain Energy / Elastic Potential Energy in Stretched Wire

1. Restoring force at elongation x: $F(x) = \frac{YAx}{L}$
2. Work done for infinitesimal elongation dx: $dW = F(x)\,dx = \frac{YAx}{L}\,dx$
3. Total work done for total extension l:

$$W = \int_0^l \frac{YAx}{L}\,dx \quad \text{Work } W = \int_0^l \frac{YA}{L}x\,dx = \frac{YA}{L}\left[\frac{x^2}{2}\right]_0^l = \frac{1}{2}\frac{YAl^2}{L} = \frac{1}{2}\left(\frac{YAl}{L}\right)l = \boxed{\frac{1}{2}F \cdot l}$$

4. Energy per unit volume (Energy Density u):

$$u = \frac{U}{\text{Volume}} \quad \text{Energy density } u = \frac{\frac{1}{2}F\,l}{A \cdot L} = \frac{1}{2}\left(\frac{F}{A}\right)\left(\frac{l}{L}\right) = \boxed{\frac{1}{2} \times \text{Stress} \times \text{Strain} = \frac{1}{2}Y(\text{Strain})^2}$$

5. Engineering Applications & Real-World Methods

Exam Oriented

BENDING OF BEAMS & MAXIMUM MOUNTAIN HEIGHT

EXAM FAVOURITE SHORTCUT

$$\text{Sag / Deflection } (\delta) = \frac{WL^3}{4bd^3Y}, \quad h_{\max} \approx \frac{\sigma_{\text{elastic}}}{\rho g} \approx 10\ \text{km}$$

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Beam Optimization: To minimize bending δ, depth d is increased rather than width b because $\delta \propto d^{-3}$. I-section beams prevent buckling and save material.

Crane Steel Rope: Minimum area required to lift load M: $A \geq \frac{Mg}{\sigma_y}$. Uses braided thin wires for flexibility and strength.

6. Standard Numerical Solving Algorithms

Step-by-Step Methods

METHOD 1: Two Wires Connected End-to-End (Series Load)

Identify Same Tension: $F_1 = F_2 = F = W$ → Express Elongations: $\Delta L_1 = \frac{FL_1}{A_1Y_1}$, $\Delta L_2 = \frac{FL_2}{A_2Y_2}$ →

Ratio Formula: $\frac{\Delta L_1}{\Delta L_2} = \left(\frac{Y_2}{Y_1}\right)\left(\frac{L_1}{L_2}\right)\left(\frac{A_2}{A_1}\right)$ → Total Elongation: $\Delta L_{\text{total}} = \Delta L_1 + \Delta L_2$

METHOD 2: Wire Whirled in a Vertical Circle (Bottom Point)

Max Tension at Lowest Point: $T_{\max} = Mg + M\omega^2 L$ → Set Stress Equation: $\sigma_{\max} = \frac{T_{\max}}{A}$ →

Calculate Elongation: $\Delta L = \frac{T_{\max} L}{AY}$

METHOD 3: Density Variation under High Hydraulic Pressure

Mass Conservation: $m = \rho V = \rho' V'$ → Volume Change: $V' = V(1 - \frac{p}{B})$ →

New Density: $\rho' = \frac{\rho}{1 - \frac{p}{B}} \approx \rho\left(1 + \frac{p}{B}\right)$

7. Common Calculation Mistakes & Exam Checklist

Error Prevention

COMMON STUDENT ERRORS & MISTAKES:

1. **Diameter vs. Radius Confusion:** $A = \pi r^2 = \pi\left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$. Forgetting to halve diameter is the #1 mistake.
2. **Unit Conversions:** $1\ \text{GPa} = 10^9\ \text{N m}^{-2}$, $1\ \text{MPa} = 10^6\ \text{N m}^{-2}$, $1\ \text{atm} = 1.013 \times 10^5\ \text{Pa}$, $1\ \text{cm}^2 = 10^{-4}\ \text{m}^2$, $1\ \text{mm}^2 = 10^{-6}\ \text{m}^2$.
3. **Tension in Mid-wire:** In a hanging wire loaded with weight W, tension at any section is W, NOT $2W$.
4. **Elasticity Misconception:** Steel is *more* elastic than rubber because steel requires a much larger stress for the same strain.